IAT New Test 10

Question paper, answer key and worked solutions

60 questions 180 minutes 240 marks

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

Instructions

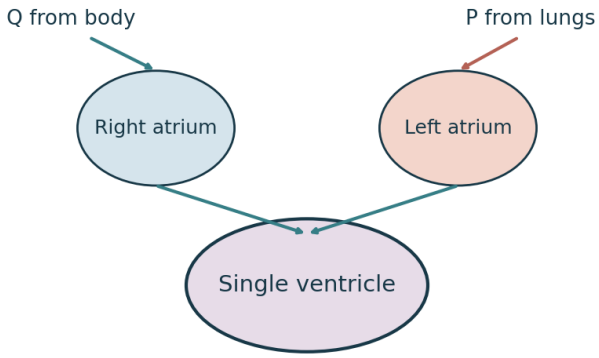
Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

Biology

Q1 A conifer bears ovules on exposed scales; after fertilisation its seeds mature without a fleshy ovary wall. Which feature most directly separates it from an angiosperm?
A) Absence of vascular tissue
B) No seed coat
C) Absence of pollen
D) No enclosed ovary forming a fruit

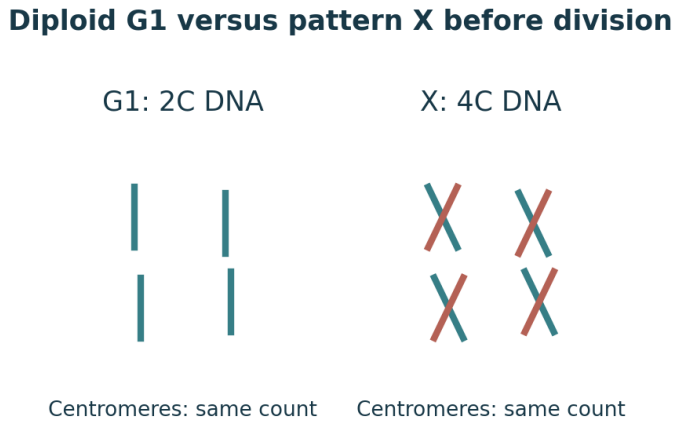
Q2 From the frog-heart figure, trace the labelled inflow vessels P and Q through the chambers. Which deduction about circulation is justified?

Frog circulation: follow the inflows



A) The ventricle receives both oxygen-rich and oxygen-poor streams
B) P brings oxygen-poor blood and Q oxygen-rich blood
C) Frogs have two fully separated ventricles
D) Blood from P bypasses the ventricle

Q3 A cultured diploid cell shows the chromosome and DNA pattern X in the figure. At which stage is this pattern expected before sister-chromatid separation?



A) G1 after cytokinesis
B) Anaphase after centromere division
C) G2 after DNA replication
D) Gamete maturation after meiosis II

Q4 Germinating oil-rich seeds consume oxygen and release less CO_2 than the volume of O_2 taken up. Which respiratory quotient is most plausible?
A) Less than 1
B) Exactly 1
C) Greater than 1
D) Undefined because seeds do not respire

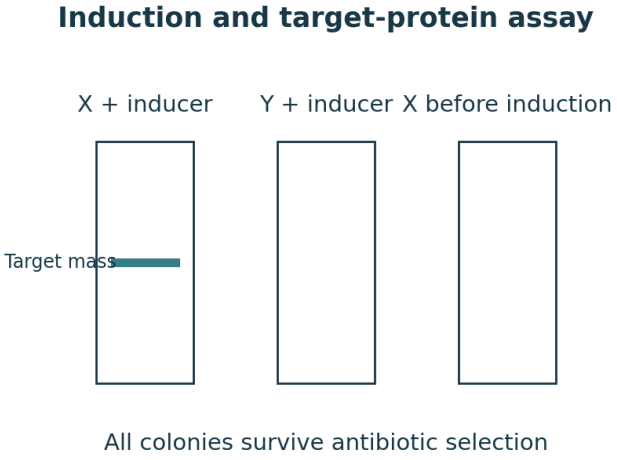
Q5 A skeletal-muscle fibre receives Ca^{2+} but a drug prevents myosin heads from binding actin. Which immediate event is blocked while the thin filament can still expose its sites?
A) ATP synthesis in mitochondria
B) Ca^{2+} binding to troponin
C) Depolarisation of the sarcolemma
D) Cross-bridge formation and force generation

Q6 A primary spermatocyte has 2n=46 chromosomes and has completed DNA replication. After meiosis I, each secondary spermatocyte has which chromosome and chromatid counts?
A) 46 chromosomes, 46 chromatids
B) 23 chromosomes, 46 chromatids
C) 23 chromosomes, 23 chromatids
D) 46 chromosomes, 92 chromatids

Q7 A coding-strand DNA triplet 5′-TAC-3′ is transcribed. Which mRNA codon and amino acid follow, using AUG=Met and UAC=Tyr?
A) 5′-UAC-3′, Tyr
B) 5′-AUG-3′, Met
C) 3′-UAC-5′, Tyr
D) 5′-TAC-3′, Tyr

Q8 A mosquito transmits infective Plasmodium sporozoites to a person. Which next location and developmental event in the human host are expected?
A) RBC division before liver entry
B) Formation of mosquito oocysts in liver
C) Immediate formation of gametes in human blood
D) Liver-cell infection and multiplication

Q9 The figure compares antibiotic-selected colonies X and Y before and after induction. Which observation establishes successful expression rather than mere plasmid uptake?



A) Both colonies' antibiotic growth
B) X's protein band at the expected mass after induction
C) Y's lack of an unrelated colony colour
D) Equal sizes of X and Y colonies

Q10 A forest reserve and adjacent farm each contain the same number of observed bird species in one survey. The reserve has multiple habitat types and many species found only there; the farm's list is mostly widespread generalists. Which conservation inference is best supported?
A) Equal observed richness proves equal biodiversity value
B) The farm must have more endemic species
C) The reserve may protect more unique lineages and habitats despite equal richness
D) Species identity never matters in biodiversity comparisons

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Q11 A mutation changes a coding DNA triplet from 5′-GAA-3′ to 5′-GAG-3′; both corresponding mRNA codons encode glutamate. What is the most specific consequence described?
A) Silent substitution at this codon
B) Frameshift at every downstream codon
C) Nonsense mutation with an early stop
D) Deletion of one amino acid

Q12 After a booster vaccine, serum antibody rises faster and higher than after the first dose, although the same antigen mass is injected. Which cellular explanation fits the time course?
A) Immediate conversion of all erythrocytes to plasma cells
B) Recall activation of antigen-specific memory B cells
C) No requirement for antigen-specific receptors
D) Only innate phagocytes can make antibody

Q13 At rest, heart rate is 70 min^{-1} and stroke volume 70 mL; effective peripheral resistance is 20 $mmHg \cdot min \cdot L^{-1}$. Later, rate is 100 min^{-1}, stroke volume 50 mL and resistance 18 in the same units. Neglect central venous pressure. Which output and mean-pressure changes follow?
A) Output falls 4.9→4.5 L/min; pressure rises 98→100 mmHg
B) Output remains 4.9 L/min; pressure remains 98 mmHg
C) Output rises 4.9→5.0 L/min; pressure falls 98→90 mmHg
D) Output rises 4.9→7.0 L/min; pressure rises 98→126 mmHg

Q14 A diploid cell has 8 chromosomes in G1. A spindle checkpoint failure lets all sister chromatids separate, but cytokinesis does not occur. Immediately after separation in that one cell, how many chromosomes and DNA molecules are present?
A) 8 chromosomes, 8 DNA molecules
B) 8 chromosomes, 16 DNA molecules
C) 16 chromosomes, 8 DNA molecules
D) 16 chromosomes, 16 DNA molecules

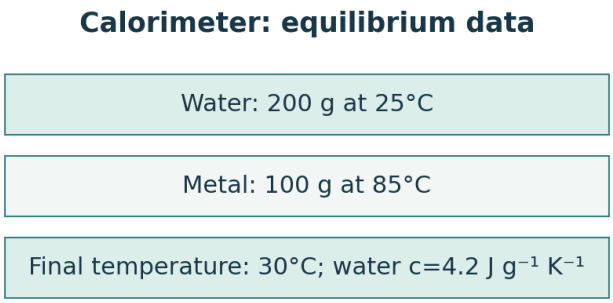
Q15 In a cycle with ovulation near day 14, progesterone is low before ovulation and rises afterward. Which structure is the main source of that post-ovulatory rise?
A) Developing follicle’s theca alone
B) Corpus luteum formed from the ruptured follicle
C) Endometrium after implantation
D) Pituitary posterior lobe

Chemistry

Q16 BF_3 has polar B–F bonds. Under its ideal trigonal-planar shape, what is the net molecular dipole moment?
A) Large and toward one F
B) Large and perpendicular to plane
C) Nonzero because each bond is polar
D) Zero because the three equal bond vectors cancel

Q17 At equal metal and oxidation state, $[M(en)_3]^{3+}$ is often more stable than a comparable complex with six separate NH_3 donors. Which explanation most directly accounts for the chelate contribution?
A) One en forms two bonds; replacing multiple free ligands can increase particle entropy
B) en always increases the metal's oxidation state
C) en has no nitrogen donor atoms
D) Monodentate complexes violate charge conservation

Q18 Using the metal-and-water calorimeter data in the figure, and neglecting heat loss and cup heat, find the metal specific heat.



A) 0.38 J g^{-1} K^{-1}
B) 0.76 J g^{-1} K^{-1}
C) 1.52 J g^{-1} K^{-1}
D) 4.20 J g^{-1} K^{-1}

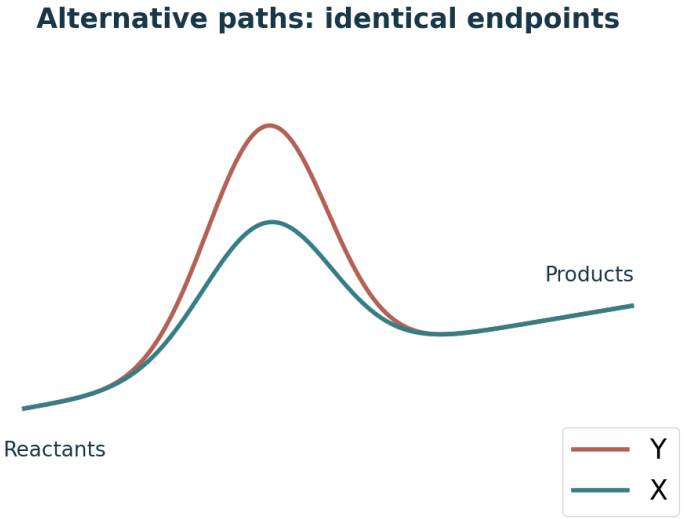
Q19 A sample contains 0.20 mol acid chloride RCOCl but only 0.16 mol methanol for a 1:1 esterification to $RCOOCH_3$. The methyl ester is isolated at 75% of theoretical yield. Of that isolated ester, 80% undergoes complete hydrolysis back to RCOOH. How many moles of acid form?
A) 0.20 mol
B) 0.12 mol
C) 0.096 mol
D) 0.080 mol

Q20 Ag^+ is deposited from solution by a current of 1.00 A for 965 s. With Faraday constant 96500 C mol^{-1}, how many moles of Ag are deposited at 100% efficiency?
A) 0.001 mol
B) 0.100 mol
C) 1.00 mol
D) 0.010 mol

Q21 Aniline is brominated with bromine water to give mainly 2,4,6-tribromoaniline. If predominantly p-bromoaniline is desired, which sequence controls the activating amino group?
A) Acetylate $–NH_2$, brominate, then hydrolyse the acyl group
B) Add excess bromine water and heat
C) Raise the pH without protecting the amino group
D) Reduce the ring before bromination

Q22 In acidic permanganate titration, MnO_4^- accepts 5 electrons per ion and Fe^{2+} donates 1. How many moles of Fe^{2+} react with 0.020 mol MnO_4^-?
A) 0.020 mol
B) 0.100 mol
C) 0.004 mol
D) 0.500 mol

Q23 The figure shows two reaction-energy profiles with the same reactants and products; path X has a lower maximum than Y. Which conclusion follows at equal temperature if both profiles represent the same overall reaction?



A) X has a more negative reaction enthalpy than Y
B) X has a higher reverse activation energy but the same forward one
C) X has a smaller forward activation barrier; ΔH is unchanged
D) X changes the equilibrium constant through a new energy gap

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Q24 In a closed vessel, an alcohol and a carboxylic acid form an ester and water reversibly. After equilibrium, removing some water without changing temperature initially does what?
A) Shifts toward reactants because water is a product
B) Leaves the reaction quotient unchanged
C) Changes the equilibrium constant
D) Shifts toward ester formation

Q25 Two compounds have molecular formula $C_4H_{10}O$. One is butan-1-ol and the other is ethoxyethane. Their relationship is best described as?
A) Functional-group isomers with alcohol versus ether
B) Geometric isomers around a double bond
C) Enantiomers with the same connectivity
D) Tautomers interconverted by proton shift

Q26 At the same temperature, compare dilute aqueous solution X containing 0.10 m glucose plus 0.05 m fully dissociated NaCl with solution Y containing 0.075 m fully dissociated $CaCl_2$. In the same solvent, what is $\Delta T_f(Y)/\Delta T_f(X)$?
A) 3/4
B) 1
C) 8/9
D) 9/8

Q27 A compound contains 40.0% C, 6.7% H and 53.3% O by mass. Its simplest whole-number formula is?
A) CH_2O
B) $C_2H_4O_2$
C) CHO
D) CH_3O

Q28 Between F and Cl, which element has the more negative first electron-gain enthalpy in the usual atomic comparison, and why?
A) F, because its atomic radius is smaller
B) Cl, because its larger 3p region reduces electron–electron crowding
C) They are equal because both are halogens
D) Neither accepts an electron exothermically

Q29 A short double-stranded DNA region has 24 base pairs, of which 9 are A–T pairs. How many G–C pairs and total hydrogen bonds does this region contain if A–T has two and G–C three bonds?
A) 15 G–C pairs; 45 bonds
B) 9 G–C pairs; 63 bonds
C) 15 G–C pairs; 63 bonds
D) 24 G–C pairs; 72 bonds

Q30 A buffer initially contains 0.20 mol HA and 0.20 mol A^- in 1 L. Add 0.05 mol strong acid with no appreciable volume change. If pKa=5.00, what pH follows?
A) 5.00
B) 5.22
C) 4.78
D) 4.52

Mathematics

Q31 A circle tangent to both coordinate axes lies in the first quadrant and passes through (4,2). Which pair gives all possible radii?
A) 2 and 10
B) 2 only
C) 4 and 8
D) 10 only

Q32 A circle of radius 5 cm has a chord 8 cm long. What is the perpendicular distance from its centre to the chord?
A) 4 cm
B) 3 cm
C) 1 cm
D) √41 cm

Q33 For f(x)=x³−3x on the closed interval [−2,2], which statement lists every location of its global maximum and minimum?
A) Max 2 only at x=2; min −2 only at x=−2
B) Max 2 at −1 only; min −2 at 1 only
C) Max 2 at −1 and 2; min −2 at −2 and 1
D) Max 2 at −2 and 1; min −2 at −1 and 2

Q34 Evaluate $\int_1^2 (x^2+1)/[x(x+1)^2], dx$.
A) ln2+1/3
B) ln2−2/3
C) ln(3/2)
D) ln2−1/3

Q35 Restrict f(x)=x²−4x+1 to x≥2. What is f⁻¹(6)?
A) 5
B) −1
C) 3
D) 6

Q36 For which real t is $\begin{pmatrix} t & 1 \\ 4 & t \end{pmatrix}$ not invertible?
A) t=2 only
B) t=−2 or 2
C) t=−4 or 4
D) No real t

Q37 An urn has 3 red and 2 blue balls. Two are drawn in order without replacement. Given that the second ball is red, what is the probability that the first was also red?
A) 3/5
B) 2/5
C) 1/2
D) 1/4

Q38 The coefficients of x^2 and x^3 in $(1+x)^n$ have ratio 3:4. What is the coefficient of x^4 in $(1+2x)^n$?
A) 60
B) 120
C) 480
D) 240

Q39 For a sequence, the sum of its first n terms is $S_n=n^2+2n$ for every $n\geq 1$. What is its tenth term?
A) 21
B) 19
C) 30
D) 120

Q40 Evaluate $\lim_{x\to 0}(e^{3x}-1)/x$.
A) 1
B) 3
C) 0
D) ∞

Q41 A line has direction vector (1,2,2), and a plane has normal $(2,-1,2)$. What is the sine of their acute angle?
A) 2/3
B) 2/9
C) 4/9
D) 8/9

Q42 How many distinct arrangements of the letters of BANANA have no two A's adjacent?
A) 6
B) 20
C) 24
D) 12

Q43 For $z=-1-\sqrt{3}i$, what is its principal argument in $(-\pi,\pi]$?
A) $-2\pi/3$
B) $2\pi/3$
C) $-\pi/3$
D) $4\pi/3$

Q44 Eliminate the arbitrary constant C from the family $y=Cx^2+1$, for $x\neq 0$. Which first-order differential equation results?
A) $xy'=y-1$
B) $xy'=2(y-1)$
C) $y'=2(y-1)$
D) $xy'=2y$

Q45 How many x in $[0,4\pi]$ satisfy $2\sin^2 x-3\sin x+1=0$ and $\cos x\geq 0$?
A) 2
B) 3
C) 4
D) 6

Physics

Q46 Two projectiles launch simultaneously from the same level: A is thrown vertically up at 20 m s^{-1}; B, from a point 30 m horizontally away, has horizontal speed 10 m s^{-1} and vertical speed 20 m s^{-1} toward A. Neglect air resistance. When do their horizontal positions first coincide, and are their heights then equal?
A) At 2 s, unequal heights
B) At 3 s, equal heights
C) At 3 s, unequal heights
D) At 4 s, equal heights

Q47 A constant horizontal force 12 N moves a cart at instantaneous velocity 3 m s^{-1} along the force. What is instantaneous mechanical power delivered?
A) 4 W
B) 15 W
C) 36 W
D) 48 W

Q48 A satellite rotates about a fixed point with negligible external torque. Its moment of inertia changes from I to I/4 while its angular speed initially is ω. What is the final angular speed?
A) ω/4
B) ω/2
C) 2ω
D) 4ω

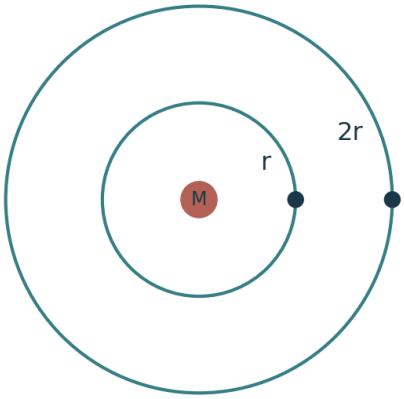
Q49 Ideal incompressible water flows from a wide lower pipe at speed 2 m s^{-1} to a narrow section 1 m higher at speed 4 m s^{-1}. With ρ=1000 kg m^{-3} and g=10 m s^{-2}, what is p_lower−p_upper?
A) 16 kPa
B) 6 kPa
C) 10 kPa
D) 22 kPa

Q50 Two different ideal gases at the same temperature and pressure are allowed to mix in an insulated rigid vessel. No chemical reaction occurs. Which statement about entropy of the combined system is correct?
A) It decreases because diffusion orders molecules
B) It increases due to additional accessible mixing arrangements
C) It remains fixed because temperature is unchanged
D) It changes sign depending only on molar mass

Q51 A string fixed at both ends has its third harmonic at 120 Hz. Keeping tension and length fixed, what is the frequency of its fifth harmonic?
A) 72 Hz
B) 160 Hz
C) 200 Hz
D) 600 Hz

Q52 Using the two circular orbital states and their radii in the figure, what is the increase in total orbital mechanical energy for the satellite?

Two circular orbital states

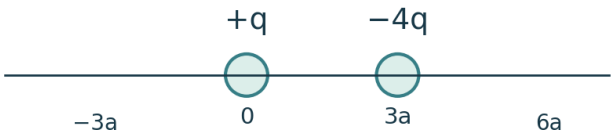


Satellite mass m, same central planet M

A) −GMm/(4r)
B) GMm/(2r)
C) −GMm/(2r)
D) GMm/(4r)

Q53 In the charge diagram, at which listed point on the x-axis is the net electric field zero?

Opposite charges on the x-axis



A) x=−3a
B) x=a
C) x=3a/2
D) x=6a

Q54 A capacitor C charged to voltage V is disconnected from its source and then connected in parallel, positive to positive, to an identical uncharged capacitor. What fraction of initial stored electrostatic energy remains in the two capacitors?
A) 1/4
B) 1/2
C) 1
D) 2

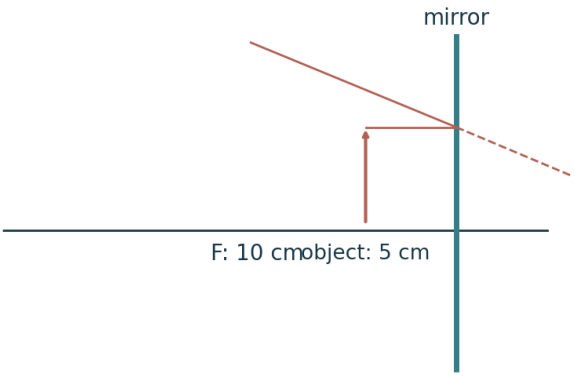
Q55 In a conductor, free-electron density is 1.0×10^{28} m^{-3} and drift-speed magnitude is 1.0×10^{-4} m s^{-1}. Taking $e=1.6\times10^{-19}$ C, what is conventional current-density magnitude?
A) 1.6×10^{3} A m^{-2}
B) 1.6×10^{4} A m^{-2}
C) 1.6×10^{5} A m^{-2}
D) 1.6×10^{6} A m^{-2}

Q56 A circular current loop's radius doubles while its current halves. What happens to magnetic-field magnitude at the centre?
A) It doubles
B) It is unchanged
C) It halves
D) It becomes one quarter

Q57 A series AC circuit contains R=30 Ω and net inductive reactance X=40 Ω. With rms supply voltage 100 V, what is mean real power?
A) 120 W
B) 200 W
C) 300 W
D) 400 W

Q58 The ray figure places an object 5 cm in front of a concave mirror with focal length 10 cm. Under the Cartesian sign convention, which image distance and nature follow?

Concave mirror: object inside focus



A) 10 cm in front, real
B) 10 cm behind, virtual
C) 5 cm behind, virtual
D) 20 cm in front, real

Q59 Two plane electromagnetic waves propagate in the same medium at equal frequency. If wave B has twice the electric-field amplitude of wave A, what is their mean energy-flux ratio IB/IA?
A) 1/2
B) 2
C) 4
D) 8

Q60 The binding-energy-per-nucleon curve peaks around intermediate mass numbers and is lower for very light nuclei. Which process involving only light nuclei can release energy according to this curve?
A) Splitting light nuclei into lighter pieces
B) Leaving nuclei unchanged
C) Fusing beyond the heavy end regardless of product
D) Fusing light nuclei toward the peak

Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

Biology

Q1 / D gymnosperm ovule exposure

Answer: (D) No enclosed ovary forming a fruit.

Why: Gymnosperm ovules are exposed on the reproductive scale, so a true ovary wall does not surround them and mature into a fruit. Seeds and pollen can occur in both groups; ovule enclosure is the decisive observation.

Why not the others: (A) conifers have vascular xylem and phloem; (B) gymnosperm seeds do form protective coats; (C) pollen is produced by male cones.

Key point: Locate the ovule relative to an ovary before classifying the seed plant.

Q2 / A frog circulatory structure

Answer: (A) The ventricle receives both streams.

Why: Gas exchange in the lungs enriches returning blood in P, whereas systemic venous blood in Q is relatively oxygen-poor. The two atria connect to one ventricle, so anatomical separation at the ventricular level is incomplete; this does not mean all outflow is uniformly mixed.

Why not the others: (B) reverses both return routes; (C) contradicts the drawn single ventricle; (D) ignores the left atrium’s outlet.

Key point: Trace inflow routes and oxygenation through the chambers.

Q3 / C chromosome versus DNA content

Answer: (C) G2 after replication.

Why: Replication doubles the DNA from 2C to 4C while each pair of sister chromatids remains joined at one centromere. Chromosome count by centromeres stays 2n in G2; the drawn pattern is not a doubling of chromosome number.

Why not the others: (A) is 2C; (B) separates centromeres so the chromosome count changes within the cell; (D) yields haploid products with different content.

Key point: Count chromosomes by centromeres and DNA by molecule content.

Q4 / A respiratory quotient

Answer: (A) Less than 1.

Why: RQ equals CO_2 evolved divided by O_2 consumed. Oxidising relatively reduced fats requires more oxygen per carbon dioxide produced than oxidising carbohydrate, matching the stated gas volumes.

Why not the others: (B) is typical of complete carbohydrate oxidation; (C) describes a different substrate or process; (D) contradicts the measured gas exchange.

Key point: Use the quotient of observed gas volumes, not just the substrate name.

Q5 / D skeletal muscle contraction

Answer: (D) Cross-bridge formation and force generation.

Why: Ca^{2+} binding moves tropomyosin away from actin's myosin-binding sites. If the myosin head itself cannot bind, the cross-bridge cannot form and a power stroke cannot transmit force, even though the upstream regulatory exposure occurs.

Why not the others: (A) is not necessarily inhibited by binding blockade; (B) is permitted in the premise; (C) is upstream and not logically excluded.

Key point: Place an inhibitor at the exact step it blocks.

Q6 / B human gametogenesis

Answer: (B) 23 chromosomes, 46 chromatids.

Why: Homologues separate in meiosis I while sister chromatids remain joined. Each daughter receives one chromosome from each homologous pair, each still made of two chromatids. The chromosome number halves but DNA/chromatid count does not halve again until meiosis II.

Why not the others: (A) incorrectly retains both homologues; (C) prematurely separates sisters; (D) describes a replicated diploid cell before the first meiotic division.

Key point: Distinguish chromosome number from chromatid number after meiosis I.

Q7 / A genetic-code reading

Answer: (A) 5′-UAC-3′, Tyr.

Why: The coding strand has the same 5′→3′ sequence as RNA apart from T replaced by U. The complementary template is read antiparallel; confusing it with the coding strand yields AUG instead. Translation reads the resulting UAC codon as tyrosine.

Why not the others: (B) complements the coding triplet; (C) reverses codon orientation; (D) retains DNA thymine in RNA.

Key point: State which strand is coding before converting a triplet to a codon.

Q8 / D pathogen life-cycle stage

Answer: (D) Liver-cell infection and multiplication.

Why: Sporozoites injected during a bite initially invade hepatocytes and undergo a pre-erythrocytic phase. Merozoites released from that phase later infect erythrocytes; the sexual cycle requires the mosquito.

Why not the others: (A) reverses the liver and blood stages; (C) confuses gametocytes with gametes and their host; (B) oocysts form in the mosquito.

Key point: Track parasite form, host and site in sequence.

Q9 / B biotechnology application evidence

Answer: (B) X's induced protein band.

Why: Antibiotic survival selects cells that carry resistance but does not show transcription or translation of the inserted gene. A band of the expected mass appearing upon induction in X, absent from control Y, supports expression of the target product. Identity may still require a specific assay.

Why not the others: (A) measures selection only; (C) uses an irrelevant phenotype; (D) colony size is not protein-specific evidence.

Key point: Include a product-specific comparison to claim expression.

Q10 / C biodiversity conservation evidence

Answer: (C) The reserve may protect more unique species and habitats.

Why: Equal counts of species do not encode turnover, rarity, endemism or habitat diversity. The extra observations favour retaining distinct assemblages in the reserve, though a single survey cannot by itself establish long-term persistence or exact population sizes.

Why not the others: (A) treats one richness number as complete evidence; (B) contradicts the observed uniqueness; (D) dismisses biologically relevant identity information.

Key point: Compare composition and habitat as well as richness.

Q11 / A mutation and disorder

Answer: (A) Silent substitution.

Why: One base is substituted without adding or deleting nucleotides, and both RNA codons specify the same amino acid. The local protein sequence remains unchanged at this position, although the sequence change could have other regulatory effects not established here.

Why not the others: (B) requires an insertion/deletion changing frame; (C) requires a stop codon; (D) requires removal of a residue.

Key point: Compare codon meanings before classifying a base substitution.

Q12 / B immune-related physiology

Answer: (B) Memory B-cell recall.

Why: The primary response builds antigen-specific memory populations. Upon re-exposure, these cells expand and differentiate efficiently into antibody-secreting cells, explaining shorter lag and larger response under the same antigen dose. The graph trend alone does not identify every antibody class.

Why not the others: (A) erythrocytes do not perform that conversion; (C) loses antigen specificity; (D) phagocytes do not secrete immunoglobulin.

Key point: Explain a secondary response through immune memory.

Q13 / C cardiac output and pressure

Answer: (C) Output rises slightly while pressure falls.

Given: Initial HR×SV=70×70=4900 mL/min; later 100×50=5000 mL/min.

Formula/Principle: Output=rate×stroke volume. Pressure≈output×resistance.

Step 1: Output changes 4.9→5.0 L/min.

Step 2: Mean pressure changes 4.9×20=98 to 5.0×18=90 mmHg.

Why: Pressure can fall despite a small rise in pumped volume if vascular resistance drops enough. The calculation links the cardiac pump and circulation rather than inferring pressure from rate alone.

Why not the others: (A) mishandles output and resistance; (B) assumes both products are fixed; (D) holds stroke volume at its earlier value.

Key point: Compute output before multiplying by the resistance at each state.

Q14 / D mitotic checkpoint

Answer: (D) 16 chromosomes, 16 DNA molecules.

Given: Eight chromosomes replicate before mitosis; the cell remains undivided after sister separation.

Formula/Principle: A separated chromatid becomes a chromosome with its own centromere and one DNA molecule.

Step 1: Eight replicated chromosomes have 16 chromatids; separation yields 16 one-chromatid chromosomes in the same cell.

Why: The checkpoint failure allows premature segregation, but it does not create new DNA beyond the S-phase copies. Because cytokinesis has not partitioned the material, both sets are counted in one cell at the specified instant.

Why not the others: (A) counts one eventual daughter; (B) uses pre-separation chromosome count; (C) mismatches chromosome and molecule counts after separation.

Key point: Specify the cell and instant when counting chromosomes.

Q15 / B menstrual hormone sequence

Answer: (B) Corpus luteum.

Why: Following LH-triggered ovulation, remnants of the follicle luteinise and secrete substantial progesterone. This hormone maintains a secretory endometrium; the temporal rise tracks formation of the corpus luteum.

Why not the others: (A) does not account for the major luteal rise; (C) is a target tissue rather than the principal ovarian source; (D) does not secrete progesterone.

Key point: Link the hormone curve to the structure present in that phase.

Chemistry

Q16 / D bond polarity versus molecular dipole

Answer: (D) Zero.

Why: Three identical bond dipoles separated by 120° in one plane have a zero vector sum. Polarity of each bond does not by itself imply a molecular dipole; geometry and symmetry determine the resultant.

Why not the others: (A) arbitrarily singles out one equivalent F; (B) no bond dipole points out of the plane; (C) adds magnitudes rather than vectors.

Key point: Sum bond dipoles as vectors.

Q17 / A chelate stability comparison

Answer: (A) Chelation and entropy.

Why: Each bidentate en binds through two nitrogens, creating a ring. Substitution of three en for six independent monodentate ligands can release more separate particles into solution, favouring complex formation entropically in a comparable system; bond strength and solvent can also matter.

Why not the others: (B) ligand denticity does not itself change oxidation state; (C) en contains two N donors; (D) charge balance can hold for either ligand set.

Key point: Compare donor connectivity and solution particle count, not only formula charge.

Q18 / B calorimetric heat balance

Answer: (B) 0.76 J g^{-1} K^{-1}.

Given: Water gains 5 K; metal loses 55 K.

Formula/Principle: Heat gained by water equals heat lost by metal.

Step 1: Water gains 200×4.2×5=4200 J.

Step 2: 100 c×55=4200, hence c=4200/5500≈0.764 J g^{-1} K^{-1}.

Why: The water and metal have different temperature changes, so using the same ΔT on each side is invalid. Heat conservation and the no-loss assumption connect the two changes.

Why not the others: (A) doubles the metal temperature loss; (C) halves it; (D) copies water's specific heat.

Key point: Use the temperature change of each body separately.

Q19 / C acid derivative transformation

Answer: (C) 0.096 mol.

Given: Methanol limits the 1:1 formation stage at 0.16 mol theoretical ester, followed by 75% isolation and 80% conversion.

Formula/Principle: Apply limiting reagent, stage yield and hydrolysis fraction in sequence.

Step 1: Isolated ester=0.16×0.75=0.12 mol.

Step 2: Acid=0.12×0.80=0.096 mol.

Why: Alcoholysis replaces chloride at the carboxylic-acid derivative to form methyl ester; hydrolysis of one ester molecule gives one acid molecule. The acid chloride excess cannot make more ester without methanol, and the second stage starts from isolated material, not theoretical production.

Why not the others: (A) uses initial acid chloride amount; (B) stops at isolated ester; (D) applies 80% to an incorrect 0.10 mol intermediate.

Key point: Identify the limiting stage before compounding yields.

Q20 / D Faraday deposition

Answer: (D) 0.010 mol.

Given: Q=It=965 C; Ag^+ requires one electron per atom.

Formula/Principle: n(Ag)=Q/(1F).

Step 1: 965/96500=0.0100 mol.

Why: Each deposited Ag consumes one mole of electrons per mole of metal. The specified efficiency permits use of the entire charge, rather than an unknown side-reaction adjustment.

Why not the others: (A) shifts the decimal one place too far; (B) shifts it in the other direction; (C) treats seconds as coulombs without applying the Faraday constant.

Key point: Convert current × time to charge before using electron stoichiometry.

Q21 / A aniline substitution control

Answer: (A) Protect, substitute, deprotect.

Why: Acetylation converts aniline to acetanilide, decreasing the extreme activation of the ring while retaining ortho/para direction. Bromination then favours monobromination, commonly para by steric preference, and hydrolysis restores $–NH_2$.

Why not the others: (B) encourages multiple substitutions; (C) does not moderate activation reliably; (D) destroys aromatic substitution context.

Key point: Temporary protection can tune reactivity without changing final functional group.

Q22 / B redox titration equivalence

Answer: (B) 0.100 mol.

Given: Electron changes are five per permanganate and one per iron ion.

Formula/Principle: Electrons gained = electrons lost at equivalence.

Step 1: $0.020 \times 5 = 0.100$ mol electrons accepted, requiring 0.100 mol Fe^{2+}.

Why: The equivalence relation uses redox electron stoichiometry rather than a 1:1 volume or molecule count. Acidic medium is specified because permanganate's reduction product and electron count depend on medium.

Why not the others: (A) assumes 1:1; (C) divides by five; (D) multiplies by 25.

Key point: Balance electron transfer before applying titration amounts.

Q23 / C rate graph and activation barrier

Answer: (C) Lower barrier, same ΔH.

Why: The barrier is the vertical rise from reactants to the transition-state maximum, which is smaller for X. Because the endpoints coincide, the enthalpy difference between products and reactants does not change. A catalyst-like lower pathway can speed the kinetics without changing equilibrium thermodynamics.

Why not the others: (A) endpoint ΔH is identical; (B) reverses the indicated forward comparison; (D) equilibrium at fixed T depends on the endpoints, not path height.

Key point: Distinguish path-dependent activation energy from state-function enthalpy.

Q24 / D esterification equilibrium

Answer: (D) Toward ester.

Why: Water is a product, so its removal lowers the product-to-reactant reaction quotient below the unchanged equilibrium constant. The forward direction generates ester and water until a new equilibrium is approached.

Why not the others: (A) reverses Le Châtelier's response; (B) removing a product alters Q; (C) K changes with temperature, held constant here.

Key point: Compare Q with K immediately after a composition change.

Q25 / A isomer classification

Answer: (A) Functional-group isomers.

Why: The two molecules share formula but differ in connectivity around oxygen: one contains an O–H bond and the other a C–O–C linkage. This changes the functional group and typical reactivity while conserving atom count.

Why not the others: (B) there is no alkene geometry; (C) connectivity is not identical; (D) an alcohol–ether pair is not a proton-shift tautomeric equilibrium.

Key point: Match formula first, then locate the functional group.

Q26 / D colligative-property comparison

Answer: (D) 9/8.

Given: Glucose contributes one particle, NaCl two, $CaCl_2$ three per formula unit under complete dissociation.

Formula/Principle: Depression in the same solvent is Kf times total effective particle molality.

Step 1: X effective molality $= 0.10 + 2(0.05) = 0.20$ m.

Step 2: Y effective molality $= 3(0.075) = 0.225$ m; ratio $0.225/0.20 = 9/8$.

Why: Colligative contributions from the two solutes in X add, whereas the three ions of Y multiply its own molality. Common Kf cancels, so neither masses nor chemical labels alone determine the

ratio. The dilute complete-dissociation premise fixes the van't Hoff factors.

Why not the others: (A) compares bare molalities; (B) assumes equal particle totals; (C) inverts the correct ratio.

Key point: Sum effective particle molality within each solution, then compare.

Q27 / A empirical formula from composition

Answer: (A) CH_2O.

Given: Imagine a 100 g sample; atomic masses C12,H1,O16.

Formula/Principle: Convert each mass to moles, then divide by the smallest.

Step 1: C=3.33, H=6.7, O=3.33 mol; ratio approximately 1:2:1.

Why: Empirical formula encodes the lowest integer atom ratio, not necessarily the molecular formula. The near equal C/O molar amounts and twice as many H atoms select CH_2O.

Why not the others: (B) is a multiple, not the simplest ratio; (C) halves hydrogen incorrectly; (D) changes both hydrogen and oxygen proportions.

Key point: Normalize moles, not mass percentages.

Q28 / B electron-affinity comparison

Answer: (B) Cl.

Why: Fluorine's compact 2p valence region creates stronger repulsion for an arriving electron; chlorine's more spacious 3p shell can accommodate it with a larger energy release. This is an exception to a simplistic size-only trend.

Why not the others: (A) ignores compact-orbital repulsion; (C) shared group does not force identical enthalpy; (D) both electron gains can be exothermic.

Key point: Orbital crowding can outweigh a monotonic periodic trend.

Q29 / C nucleic-acid base pairing

Answer: (C) 15 pairs; 63 bonds.

Given: Twenty-four pairs in all, nine A–T pairs.

Formula/Principle: G–C pairs=24−9; bonds=2nAT+3nGC.

Step 1: 15 G–C pairs and 2×9+3×15=18+45=63 hydrogen bonds.

Why: Each counted pair is one rung of the duplex, regardless of strand orientation. Multiplying all 24 by three overcounts A–T bonds, while counting bases rather than base pairs doubles the rung count.

Why not the others: (A) omits A–T bonds; (B) miscounts G–C; (D) assigns every pair three bonds.

Key point: Separate base-pair counts from bond counts.

Q30 / C buffer before and after neutralisation

Answer: (C) 4.78.

Given: Strong acid consumes 0.05 mol A^- and produces 0.05 mol HA.

Formula/Principle: pH=pKa+log([A^-]/[HA]).

Step 1: New mole amounts A^-=0.15, HA=0.25; ratio 0.6.

Step 2: pH=5+log0.6≈5−0.222=4.78.

Why: Neutralisation must be accounted for before applying the buffer relation. The common volume cancels in the concentration ratio, and acid addition lowers pH as expected.

Why not the others: (A) ignores added acid; (B) reverses the ratio; (D) applies too large a change.

Key point: Do the stoichiometric reaction first, then equilibrium ratio.

Mathematics

Q31 / A circle tangency

Answer: (A) 2 and 10.

Given: Tangencies require centre (r,r) and radius r>0.

Formula/Principle: $(4-r)^2+(2-r)^2=r^2$.

Step 1: Simplify to $r^2-12r+20=0$.

Worked derivation:

$$r = \frac{12 \pm \sqrt{144-80}}{2} = \frac{12 \pm 8}{2} \in \{2, 10\}.$$

Why: Both positive roots define a first-quadrant circle tangent to both axes; the specified point lies on each. Discarding the larger circle simply because its centre is farther away imposes an unstated size limit.

Why not the others: (B) loses the larger valid root; (C) does not solve the point equation; (D) loses the smaller root.

Key point: Check all geometric solutions of the tangency constraint.

Q32 / B chord distance

Answer: (B) 3 cm.

Given: Half-chord=4 cm, radius=5 cm.

Formula/Principle: The centre-to-midpoint perpendicular forms a right triangle.

Step 1: $d^2+4^2=5^2$.

Worked derivation:

$$d = \sqrt{5^2-(8/2)^2} = \sqrt{25-16} = 3\,\text{cm}.$$

Why: The perpendicular bisects a chord, so the relevant triangle uses half its length, not the full eight centimetres. The positive root represents a distance.

Why not the others: (A) takes half-chord as distance; (C) subtracts lengths; (D) adds squared lengths.

Key point: Bisect the chord before using Pythagoras.

Q33 / C extreme value with endpoints

Answer: (C) Both extrema are attained twice.

Given: Closed interval with two stationary points and two endpoints.

Formula/Principle: $f'(x)=3(x^2-1)$; evaluate f at $-2,-1,1,2$.

Step 1: Values are $-2,2,-2,2$ respectively.

Worked derivation:

$$f(-2)=-2,\ f(-1)=2,\ f(1)=-2,\ f(2)=2; \quad \max f = 2 \text{ at } \{-1,2\},\ \min f = -2 \text{ at } \{-2,1\}.$$

Why: A derivative sign chart alone gives local behaviour but cannot exclude equal values at the closed endpoints. Testing all four candidates reveals a tie for each global extremum.

Why not the others: (A) omits stationary ties; (B) omits endpoint ties; (D) exchanges maximum and minimum locations.

Key point: Compare every critical point and endpoint, preserving ties.

Q34 / D partial fractions with limits

Answer: (D) ln2−1/3.

Given: The denominator has a repeated factor (x+1)², unlike distinct-pole partial fractions.

Formula/Principle: (x²+1)/[x(x+1)²]=1/x−2/(x+1)².

Step 1: Integrate logarithmic and inverse-linear components separately.

Worked derivation:

$$\int_1^2 \left(\frac{1}{x} - \frac{2}{(x+1)^2}\right) dx = \left[\ln x + \frac{2}{x+1}\right]_1^2 = \ln 2 + \frac{2}{3} - 1 = \ln 2 - \frac{1}{3}.$$

Why: Solving the repeated-pole decomposition yields a zero coefficient for 1/(x+1) but a nonzero second-order term. Its antiderivative is rational, not a second logarithm.

Why not the others: (A) reverses the rational endpoint difference; (B) doubles that difference; (C) ignores the repeated-pole contribution.

Key point: A repeated factor needs its own higher-power partial fraction.

Q35 / A inverse restriction

Answer: (A) 5.

Given: f(x)=(x−2)²−3, branch x≥2.

Formula/Principle: Set 6=(x−2)²−3 and select the allowed root.

Step 1: (x−2)²=9, giving x=5 or −1.

Worked derivation:

$$f^{-1}(6) = 2 + \sqrt{6+3} = 5, \qquad x \geq 2.$$

Why: The restriction makes the parabola one-to-one by selecting its increasing branch. The other algebraic preimage is outside the domain of the inverse and cannot be used.

Why not the others: (B) is the excluded preimage; (C) misreads the square root; (D) confuses input with output.

Key point: An inverse of a restricted quadratic uses one branch.

Q36 / B inverse existence

Answer: (B) −2 or 2.

Given: A 2×2 matrix with parameter on both diagonal entries.

Formula/Principle: Invertibility fails exactly when determinant is zero.

Step 1: det=t²−4=(t−2)(t+2).

Worked derivation:

$$\det A = t^2 - (1)(4) = 0 \iff t \in \{-2, 2\}.$$

Why: Both signs of t make the two rows dependent, even though the off-diagonal numbers are positive. Checking only t=2 loses a genuine singular matrix.

Why not the others: (A) omits −2; (C) treats the product four as the root magnitude; (D) ignores the determinant zeros.

Key point: Factor the determinant before deciding inverse existence.

Q37 / C without-replacement conditional event

Answer: (C) 1/2.

Given: Conditional observation is the second draw red, not at least one red.

Formula/Principle: $P(R_1|R_2)=P(R_1\cap R_2)/P(R_2)$, with $P(R_2)=3/5$ by symmetry.

Step 1: $P(R_1\cap R_2)=(3/5)(2/4)=3/10$.

Worked derivation:

$$P(R_1 \mid R_2) = \frac{(3/5)(2/4)}{3/5} = \frac{1}{2}.$$

Why: Observing a red second ball means one original red has been used, leaving two red among four balls for the first position under reverse sampling symmetry. The conditional denominator is not the event that either draw is red.

Why not the others: (A) repeats the marginal; (B) reports the blue share; (D) multiplies fractions without normalising by $P(R_2)$.

Key point: Condition on the specified position, not a colour appearing anywhere.

Q38 / D neighboring-coefficient identity

Answer: (D) 240.

Given: $C(n,2)/C(n,3)=3/(n-2)$.

Formula/Principle: Solve $3/(n-2)=3/4$, then coefficient $x^4=C(n,4)2^4$.

Step 1: $n-2=4$, so $n=6$.

Worked derivation:

$$\frac{\binom{n}{2}}{\binom{n}{3}} = \frac{3}{n-2} = \frac{3}{4} \Rightarrow n = 6; \quad [x^4](1+2x)^6 = \binom{6}{4}2^4 = 15 \cdot 16 = 240.$$

Why: The ratio determines the integer exponent before the second binomial expansion is evaluated. The coefficient in $(1+2x)^n$ includes a power of two that is absent from the initial ratio.

Why not the others: (A) ignores the 2^4 factor partly; (B) halves the resulting coefficient; (C) doubles it.

Key point: Distinguish binomial coefficients from full polynomial coefficients.

Q39 / A partial-sum inversion

Answer: (A) 21.

Given: An explicit formula for partial sums, not for individual terms.

Formula/Principle: $a_n=S_n-S_{n-1}$.

Step 1: $S_{10}=120$ and $S_9=99$.

Worked derivation:

$$a_{10} = S_{10} - S_9 = (10^2 + 20) - (9^2 + 18) = 120 - 99 = 21.$$

Why: The nth term is the new contribution when moving from n−1 to n; substituting n=10 into S directly gives all ten terms together.

Why not the others: (B) is an off-by-one arithmetic result; (C) uses a naive 3n pattern; (D) reports S_{10}.

Key point: Invert a partial-sum formula by taking consecutive differences.

Q40 / B exponential limit

Answer: (B) 3.

Given: Exponential approaches one, yielding an indeterminate 0/0 form.

Formula/Principle: lim(u→0)(e^u−1)/u=1.

Step 1: Set u=3x; the expression is 3(e^u−1)/u.

Worked derivation:

$$\lim_{x\to 0}\frac{e^{3x}-1}{x}=3\lim_{u\to 0}\frac{e^{u}-1}{u}=3.$$

Why: The chain factor three survives when converting the denominator from x to 3x. It equals the derivative of e^{3x} at zero. A direct substitution gives 0/0 and therefore cannot determine the value by itself.

Why not the others: (A) omits the chain factor; (C) substitutes directly into an indeterminate ratio; (D) treats 0/0 as divergence.

Key point: Rescale the standard exponential limit correctly.

Q41 / C line-plane angle

Answer: (C) 4/9.

Given: Dot product of direction and plane normal is 4; both vector magnitudes are 3.

Formula/Principle: sin(angle between line and plane)=|d·n|/(|d||n|).

Step 1: 4/(3×3)=4/9.

Worked derivation:

$$\sin\alpha=\frac{|(1,2,2)\cdot(2,-1,2)|}{\sqrt{1+4+4}\sqrt{4+1+4}}=\frac{|2-2+4|}{9}=\frac{4}{9}.$$

Why: The dot product gives cosine of the line–normal angle, equal to sine of the complementary line–plane angle. Using cosine directly for the requested angle would swap the geometry.

Why not the others: (A) adds components; (B) halves the dot product; (D) doubles it.

Key point: A plane angle is complementary to its normal angle.

Q42 / D identical-item grouping

Answer: (D) 12.

Given: Non-A letters B,N,N have 3!/2!=3 orders.

Formula/Principle: Insert three identical A's into three of four gaps around the ordered non-A letters.

Step 1: C(4,3)=4 gap choices per non-A order.

Worked derivation:

$$N=\frac{3!}{2!}\binom{4}{3}=3\cdot 4=12.$$

Why: A gap accepts at most one A, enforcing separation without overcounting the identical A's or N's. Rotations are distinct here because the word is written in a line.

Why not the others: (A) misses two gap patterns; (B) counts some adjacent A's; (C) treats repeated N's as distinct.

Key point: Arrange the other letters first, then choose occupied gaps.

Q43 / A argument branch

Answer: (A) −2π/3.

Given: z lies in quadrant III with reference angle π/3.

Formula/Principle: Select the coterminal angle inside the stated principal range.

Step 1: The geometric angle is 4π/3; subtract 2π.

Worked derivation:

$$\operatorname{Arg}(-1-\sqrt{3}i)=\frac{4\pi}{3}-2\pi=-\frac{2\pi}{3}.$$

Why: A principal argument is a representative of a direction within a specified interval, not any coterminal angle. Negative real and imaginary parts fix the quadrant before branch reduction.

Why not the others: (B) is above the real axis; (C) has the wrong reference direction; (D) is coterminal but outside the principal range.

Key point: Check both quadrant and branch convention.

Q44 / B formation from curve family

Answer: (B) xy′=2(y−1).

Given: y−1=Cx² and y′=2Cx.

Formula/Principle: Replace C=(y−1)/x² in the derivative.

Step 1: y′=2(y−1)/x.

Worked derivation:

$$y'=2Cx=2\frac{y-1}{x^2}x \quad\Longrightarrow\quad xy'=2(y-1).$$

Why: Differentiation of the one-parameter family supplies one equation, and elimination removes the arbitrary constant. The domain x≠0 permits division; at x=0 every original curve meets y=1.

Why not the others: (A) loses the factor two; (C) omits x in the denominator; (D) omits the fixed +1 shift.

Key point: Differentiate and eliminate the parameter, retaining domain assumptions.

Q45 / C periodic solution count

Answer: (C) Four.

Given: Factor (2sin x−1)(sin x−1)=0.

Formula/Principle: Count both branches on the closed two-period interval, then filter by cos x≥0.

Step 1: sin x=1 at π/2,5π/2, both allowed; sin x=1/2 at π/6,5π/6,13π/6,17π/6, of which π/6,13π/6 pass.

Worked derivation:

$$x\in\{\pi/2,5\pi/2,\pi/6,13\pi/6\};\qquad N=4.$$

Why: The inequality rejects the second-quadrant roots but retains points where cosine is zero. Neither endpoint 0 nor 4π solves the original quadratic, so the closed interval adds no boundary root.

Why not the others: (A) counts only one branch; (B) drops a valid zero-cosine root; (D) counts both disallowed second-quadrant roots.

Key point: Solve and filter each trigonometric branch on the stated interval.

Physics

Q46 / B projectile interception

Answer: (B) At 3 s, equal heights.

Given: Horizontal gap 30 m; B's horizontal speed 10 m s⁻¹; vertical initial velocities equal.

Formula/Principle: Relative horizontal position closes at 10t=30; vertical displacements are both 20t−gt²/2.

Step 1: t=3 s; at that instant the vertical coordinates are equal for any common g.

Worked derivation:

$$x_B - x_A = 30 - 10t = 0 \Rightarrow t = 3\,\text{s}; \quad y_A = y_B = 20t - \tfrac{1}{2}gt^2.$$

Why: Equal vertical initial velocities and equal gravitational acceleration maintain equal height throughout the flight, so only horizontal closure needs solving. At g≈10, both are 15 m above launch level at 3 s.

Why not the others: (A) closes only 20 m; (C) ignores the identical vertical motion; (D) passes horizontal alignment.

Key point: Separate independent horizontal and vertical motions.

Q47 / C power at instant

Answer: (C) 36 W.

Given: Force and velocity are parallel.

Formula/Principle: P=F·v=Fv cosθ.

Step 1: θ=0 and P=12×3=36 W.

Worked derivation:

$$P = \mathbf{F}\cdot\mathbf{v} = (12)(3)\cos 0 = 36\,\text{W}.$$

Why: Power is work per unit time at the current velocity, so the distance travelled so far is not needed. A perpendicular force would deliver zero instantaneous power even with the same magnitude.

Why not the others: (A) divides force by speed; (B) adds magnitudes; (D) uses a wrong product.

Key point: Use the force component along velocity.

Q48 / D angular momentum with changing radius

Answer: (D) 4ω.

Given: External torque is negligible.

Formula/Principle: Angular momentum L=Iω remains constant.

Step 1: Iω=(I/4)ωf.

Worked derivation:

$$\omega_f = \frac{I}{I/4}\omega = 4\omega.$$

Why: Pulling mass closer to the rotation axis decreases inertia, so angular speed rises to conserve momentum. Rotational kinetic energy may change because internal work is done; it need not be constant.

Why not the others: (A) reverses the ratio; (B) halves rather than quarters inertia; (C) misses a factor two.

Key point: Conserve angular momentum, not automatically kinetic energy.

Q49 / A Bernoulli with elevation

Answer: (A) 16 kPa.

Given: Speeds 2,4 m s^{-1} and elevation difference +1 m.

Formula/Principle: Bernoulli p+½ρv²+ρgh=constant along streamline.

Step 1: Pressure difference=½ρ(4²−2²)+ρg(1)=6000+10000 Pa.

Worked derivation:

$$p_l - p_u = \tfrac{1}{2}(1000)(16-4) + (1000)(10)(1) = 16000\,\text{Pa} = 16\,\text{kPa}.$$

Why: Pressure pays for both the kinetic-energy increase and the elevation gain. Using only one of those terms gives an incomplete drop. The higher outlet has both greater gravitational potential per unit volume and greater kinetic energy per unit volume.

Why not the others: (B) omits elevation; (C) omits speed; (D) adds the lower kinetic energy twice.

Key point: Retain both height and speed changes in Bernoulli's equation.

Q50 / B entropy in mixing

Answer: (B) It increases.

Why: Removing the partition gives each gas access to more spatial arrangements while total energy is unchanged in the ideal-gas approximation. For distinct gases, the additional mixing microstates raise entropy even when initial temperatures and pressures agree.

Why not the others: (A) reverses accessible-state reasoning; (C) entropy depends on accessible configurations as well as temperature; (D) molar mass alone does not set the sign.

Worked derivation:

$$\Delta S_{\text{mix}} = -R(n_A \ln x_A + n_B \ln x_B) > 0 \quad \text{for } 0 < x_A, x_B < 1.$$

Key point: Equal temperature does not imply zero entropy of mixing.

Q51 / C standing-wave node shift

Answer: (C) 200 Hz.

Given: $f_3 = 3f_1 = 120$ Hz.

Formula/Principle: Allowed fixed-end frequencies $f_n = n\,v/(2L)$.

Step 1: $f_1 = 40$ Hz; $f_5 = 5 \times 40 = 200$ Hz.

Worked derivation:

$$f_5 = \frac{5}{3} f_3 = \frac{5}{3}(120) = 200\,\text{Hz}.$$

Why: Changing the harmonic number changes the number of antinodes while wave speed and string length remain unchanged. Node positions shift, but the fundamental spacing remains 40 Hz.

Why not the others: (A) inverses the ratio; (B) uses fourth harmonic; (D) multiplies by five without dividing by three.

Key point: Harmonic frequencies scale with their integer mode number.

Q52 / D orbital energy transfer

Answer: (D) GMm/(4r).

Given: Initial/final circular radii r and 2r.

Formula/Principle: Circular-orbit total energy E(R)=−GMm/(2R).

Step 1: Ei=−GMm/(2r), Ef=−GMm/(4r).

Worked derivation:

$$\Delta E = E_f - E_i = -\frac{GMm}{4r} + \frac{GMm}{2r} = \frac{GMm}{4r}.$$

Why: The higher orbit is less tightly bound, so its total energy is less negative and must increase. The kinetic energy itself decreases, demonstrating why kinetic energy alone cannot be used as transfer cost.

Why not the others: (A) reverses final minus initial; (B) uses the full initial binding magnitude; (C) gives a negative twice-as-large change.

Key point: Use total circular-orbit energy for each endpoint.

Q53 / A field cancellation locus

Answer: (A) x=−3a.

Given: Opposite-sign charges at 0 and 3a, magnitudes q and 4q.

Formula/Principle: Fields reinforce between opposite charges. Outside, equate magnitudes.

Step 1: Left of 0 let distance to +q be s; distance to −4q is s+3a. Then s+3a=2s.

Worked derivation:

$$s = 3a, \qquad x = -s = -3a.$$

Why: Cancellation can occur only outside the interval on the side of the smaller charge. At x=−3a the −4q is twice as far and its fourfold charge is exactly offset by inverse-square distance.

Why not the others: (B) and (C) lie between charges where fields reinforce; (D) is outside on the stronger-charge side where equality is impossible.

Key point: Establish field directions before equating magnitudes.

Q54 / B capacitor energy after reconnection

Answer: (B) One half.

Given: Initial charge Q=CV; total final capacitance 2C at fixed net Q.

Formula/Principle: Ui=Q²/(2C); Uf=Q²/[2(2C)].

Step 1: Uf/Ui=1/2.

Worked derivation:

$$\frac{U_f}{U_i} = \frac{Q^2/(4C)}{Q^2/(2C)} = \frac{1}{2}.$$

Why: The shared final voltage is V/2. Charge is conserved across isolated conductors, while energy decreases as currents dissipate energy in the connecting circuit; it is not conserved as electrostatic storage.

Why not the others: (A) squares voltage without counting doubled capacitance; (C) assumes storage energy conservation; (D) assumes voltage stays fixed by a battery.

Key point: Choose the invariant charge after disconnection.

Q55 / C drift versus current density

Answer: (C) 1.6×10^{5} A m^{-2}.

Given: $n=10^{28}$ m^{-3}, $e=1.6\times10^{-19}$ C, $v_d=10^{-4}$ m s^{-1}.

Formula/Principle: $|J|=nev_d$.

Step 1: Powers sum $28-19-4=5$.

Worked derivation:

$$|J| = (10^{28})(1.6\times10^{-19})(10^{-4}) = 1.6\times10^{5}\,\mathrm{A\,m^{-2}}.$$

Why: Electron drift direction is opposite conventional current, but the requested magnitude is positive. Multiplying density, charge and speed also produces the correct A/m² dimension.

Why not the others: (A) and (B) lose powers of ten; (D) adds an extra factor ten.

Key point: Use number density, not total electron count.

Q56 / D field of current loop

Answer: (D) One quarter.

Why: The centre field for a circular loop scales as I/R. Halving current supplies a factor 1/2 and doubling radius another 1/2, making 1/4 overall, with the current direction unchanged.

Why not the others: (A) reverses radius dependence; (B) assumes factors cancel; (C) accounts for only one of the two changes.

Worked derivation:

$$B = \frac{\mu_0 I}{2R}, \quad \frac{B'}{B} = \frac{(I/2)/(2R)}{I/R} = \frac{1}{4}.$$

Key point: Compare proportional factors before substituting constants.

Q57 / A rms real power

Answer: (A) 120 W.

Given: Impedance magnitude $\sqrt{30^2+40^2}=50\ \Omega$.

Formula/Principle: $I_{rms}=V_{rms}/Z$; $P=I_{rms}^2R=V_{rms}I_{rms}\cos\varphi$.

Step 1: $I_{rms}=100/50=2$ A; $\cos\varphi=30/50=0.6$.

Worked derivation:

$$P = I_{\mathrm{rms}}^2 R = (2)^2(30) = 120\,\mathrm{W} = 100(2)(0.6).$$

Why: The reactive voltage component affects current and phase but does not consume average real power in ideal reactance. Using V^2/R as though the entire supply were across the resistor would greatly overstate power.

Why not the others: (B) uses apparent power VI; (C) uses a wrong current; (D) squares supply voltage with an incorrect impedance.

Key point: Compute impedance and current before resistive power.

Q58 / B real-virtual image sign

Answer: (B) 10 cm behind, virtual.

Given: $u=-5$ cm and $f=-10$ cm.

Formula/Principle: Mirror equation $1/f=1/v+1/u$.

Step 1: $1/v=-1/10+1/5=+1/10$, hence $v=+10$ cm.

Worked derivation:

$$v = \left(\frac{1}{f}-\frac{1}{u}\right)^{-1} = \left(-\frac{1}{10}+\frac{1}{5}\right)^{-1} = +10\,\mathrm{cm}.$$

Why: Object inside the focal length produces diverging reflected rays whose backward extensions meet behind the mirror. Positive v in this convention means a virtual image on the far side.

Why not the others: (A) reverses distance sign; (C) equates object and image distance; (D) places a real image where reflected rays do not meet.

Key point: Interpret sign together with the ray geometry.

Q59 / C energy flux comparison

Answer: (C) Four.

Why: In the same medium, time-averaged intensity is proportional to the square of electric-field amplitude. Doubling the amplitude makes the energy transported per unit area per unit time four times greater; frequency equality avoids an unrelated comparison.

Why not the others: (A) inverses the trend; (B) assumes linear scaling; (D) cubes the amplitude ratio.

Worked derivation:

$$\frac{I_B}{I_A} = \left(\frac{E_B}{E_A}\right)^2 = 2^2 = 4.$$

Key point: Energy flux follows field amplitude squared.

Q60 / D binding-energy per nucleon

Answer: (D) Fusing toward the peak.

Why: Greater binding energy per nucleon means the final nucleons are more tightly bound and the total mass-energy is lower. Fusion of sufficiently light nuclei toward the intermediate peak can therefore release the mass difference as energy.

Why not the others: (A) generally moves away from stronger binding for light nuclei; (B) produces no binding change; (C) asserts release without comparing initial and final binding.

Worked derivation:

$$Q = B_{\text{final}} - B_{\text{initial}} > 0 \quad \text{when total final binding increases.}$$

Key point: Compare total binding, not simply nuclear mass number.
